

Goodness Ayinmode

Davis, CA | goayinmode@ucdavis.edu | Google Scholar | Website

EDUCATION

University of California, Davis

Davis, CA

PhD Student, Computer Science Advisor: Prof. Cindy Rubio-González

2023–Present

- Researching and building analysis and testing tools for software reliability, automated bug detection, and program optimization in software systems.

University of Ibadan

Ibadan, Nigeria

B.Sc. (Hons) Computer Science — First Class Honors

2018–2021

- Final-year dissertation — *UI Energy*: a Flutter/Laravel mobile app letting University of Ibadan residents buy electricity units online via a secure payment gateway, replacing in-person bank and vendor visits. [Publication]
- Relevant Coursework: Applied Linear Algebra, Machine Learning, Distributed Database Systems, Data Structures and Algorithms.

PUBLICATIONS

*Understanding and Detecting Scalability Faults in Large-Scale Distributed Systems**

IEEE TSE (accepted) [Paper] [Code]

A static and dynamic analysis tool for detecting scalability faults, faults that stay latent under small-scale testing but emerge only under large-scale deployment. It traces these faults to code anti-patterns tied to iterative code fragments whose iteration count scales with data, load, or cluster size.

Automated Bug Reproduction in Large-Scale Distributed Systems¹

Submitted

Built an LLM-driven scalability bug reproduction framework that takes issue reports and buggy codebases to synthesize bug-reproducing workloads, evaluating success with a symptom-driven oracle instead of the traditional fail-pass oracle.

*Code-Level Scalability Issues of LLM Serving Infrastructures**

Submitted

Empirical study of code-level scalability faults across LLM serving systems like Ray Serve, vLLM, and SGLang, mining 50,000+ issue reports to derive a taxonomy of real-world faults spanning the control-plane and data-plane components of the serving software stack.

ONGOING RESEARCH

Leveraging LLMs to Repair Scalability Bugs in Distributed Systems

2026–Present

Repairing scalability faults using coding agents. The challenge is not in code generation, but in evaluating a patch and extracting and utilizing the right feedback to guide patch refinement.

ADDITIONAL RESEARCH PROJECTS

Graph-Based Learning for Scalability-Fault Code Patterns

Prototyped a tool that converts Java methods into code graphs and embeds them as vectors, testing whether known *dimensional code fragments* cluster together in embedding space as a possible signal for detecting unknown ones.

EXPERIENCE

Google Summer of Code — Open Source Program Office (OSPO)

2023

Open Source Research Contributor

Remote

- Built a dataset of reproducible scalability bugs for distributed systems by analyzing bug reports, designing triggering workloads, and validating reproducibility in Docker. [Report]

Carrot Credit

2022–2023

Software Engineering Intern

Lagos, Nigeria

- Fixed production bugs and implemented UI improvements across web applications built with React and Node.js.

TECHNICAL SKILLS

Programming Languages: Python, Java, SQL, JavaScript, Go, Prolog

Tools: CodeQL, LLM coding agents (OpenAI Codex, Claude Code), Graph2Vec, FAISS, Docker, Linux, Git

TEACHING

Teaching Assistant, ECS 140A: Programming Languages

Spring 2025, Fall 2025

University of California, Davis — led discussions and office hours

Davis, CA

¹First author. *Co-first author (equal contribution).